

Sea Scallop TOR-5

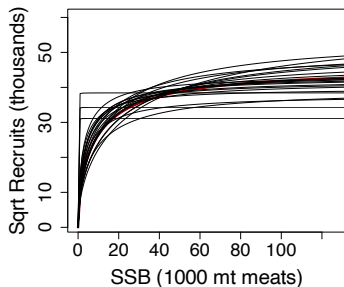
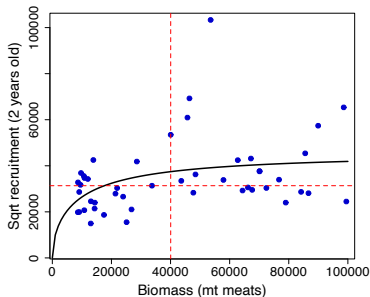
Update or redefine Status Determination Criteria (SDC; point estimates or proxies for BMSY, BTHRESHOLD, FMY and MSY reference points) and provide estimates of those criteria and uncertainty, along with a description of the sources of uncertainty. If analytic model-based estimates are unavailable, consider recommending alternative measurable proxies for reference points. Compare estimates of current stock size and fishing mortality to existing, and any redefined, SDCs. Provide stock status based on updated reference points.

- For each run, estimates yield curve Y using the equations $Y = yR$ and $R = s(bR)$, where y is yield per recruit, R is recruitment, s is the stock-recruit relationship, and b is biomass per recruit (Beverton & Holt 1957).
- Takes into account parameter uncertain by Monte-Carlo simulation - each simulation draws parameters from estimated distributions, including covariance with other parameters
- Uses trimmed mean yield and biomass curves for aggregated estimates: for each F , this is the mean yield or biomass over all the runs (100,000 in our case), with the top and bottom 10% excluded.

Georges Bank Stock-Recruit Relationship

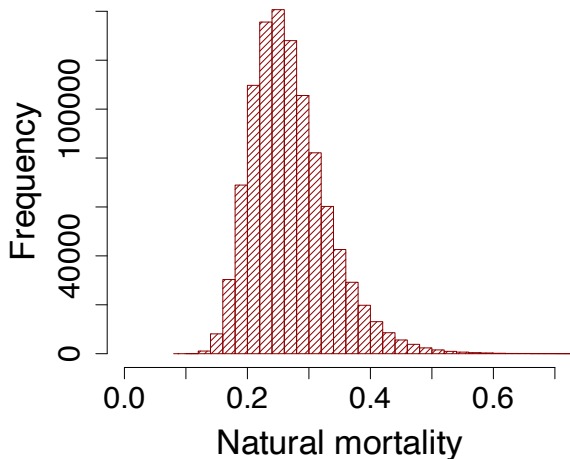
Recruits from the Georges Bank Open and Closed CASA model were advanced to age 2 using CASA estimated M s, and then recruits and biomass from the two models were summed.

Beverton-Holt S-R relationship estimated (including parameter standard errors) using square root transformed recruitment

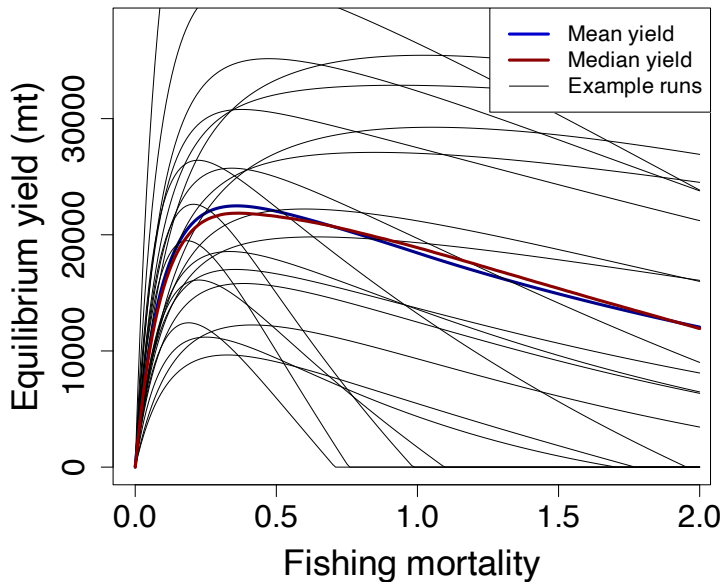


Georges Bank Natural Mortality

Simulated as $1/\gamma$ with $\mu = 0.27$ and $\sigma = 0.067$

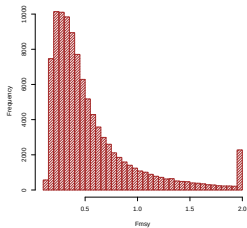


Georges Bank Yield Curves

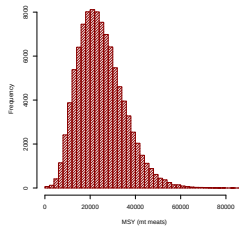


GB Reference Point Distributions

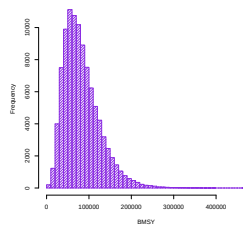
F_{MSY}



MSY



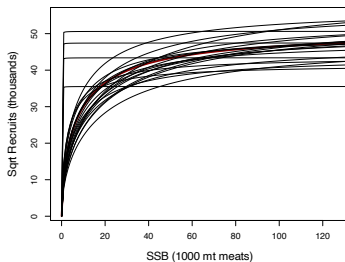
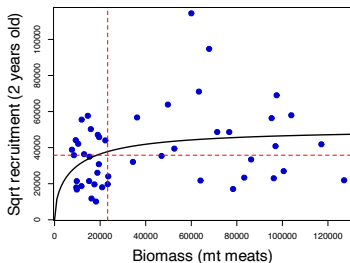
B_{MSY}



Mid-Atlantic Stock-Recruit Relationship

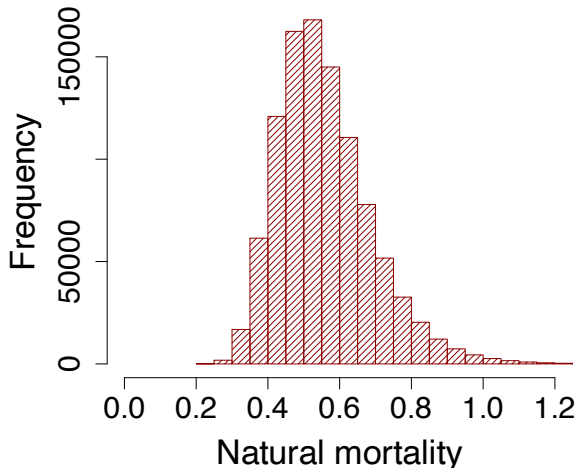
Recruitment from the last five years was on average 59% below the stock-recruit predictions. For the baseline Mid-Atlantic run, recruitment as predicted by the stock-recruit relationship was reduced by 59%.

Beverton-Holt S-R relationship fit using square root transformed recruitment.



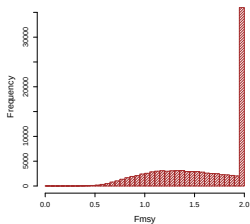
Mid-Atlantic Natural Mortality

Used a mean of $M = 0.56$, based on the adult natural mortality in the last five years in the CASA model.

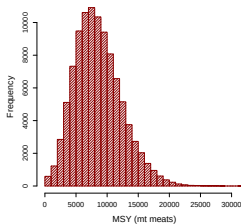


Mid-Atlantic Reference Point Distributions

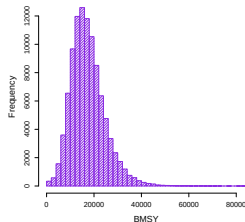
F_{MSY}



MSY

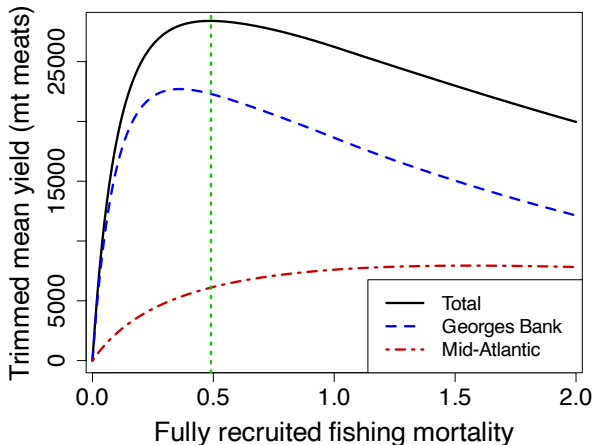


B_{MSY}



Trimmed mean yield curves

Mid-Atlantic: $F_{MSY} = 1.56$ $B_{MSY} = 15909$ $MSY = 7941$
Georges Bank: $F_{MSY} = 0.36$ $B_{MSY} = 83414$ $MSY = 22706$
Combined: $F_{MSY} = 0.49$ $B_{MSY} = 93282$ $MSY = 28402$



Stock Biomass and Status

Combined biomass below target B but above threshold, so not overfished.

$F_{2023} = 0.33 < F_{MSY} = 0.49$ so no overfishing.

However, GB $F_{2023} = 0.47$ is above its $F_{MSY} = 0.36$.

